Alternative Assessment Tool (AAT) Report on

"Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL"

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)

[Course Code: 23ECSPE1AD]

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INTRODUCTION

Cadence is an Electronic Design Automation (EDA) tool used for designing, simulating, and verifying digital circuits. It helps in understanding the behaviour and functionality of digital systems before implementing them in hardware.

In this assignment, Cadence software is used to design and simulate two digital circuits: a 4-bit Comparator and a Ring Counter. The 4-bit Comparator is a combinational logic circuit that compares two 4-bit binary numbers and determines whether one number is greater than, less than, or equal to the other. The Ring Counter is a sequential logic circuit that uses flip-flops connected in a circular arrangement to generate a sequence of binary states.

The simulation helps verify the logical operation of both circuits by observing their outputs for different input combinations and clock pulses. This assignment provides practical knowledge of combinational and sequential circuit design, simulation, and verification using Cadence software.

COMBINATIONAL CIRCUIT

1. FOUR-BIT COMPARATOR

A 4-bit comparator is a combinational circuit that compares two unsigned 4-bit binary numbers, A and B. It produces three mutually exclusive outputs: A greater than B, A less than B, or A equal to B.

PROCEDURE

1. Create folders with names “RTL”, “TB” and “RESULT” to store the code, testbench and simulation results respectively.
2. Open the terminal.
3. Open the folder RTL using cd command.
4. Use “gedit 4bit_comparator.v” to open and edit a Verilog file, add your code her and save it.
5. To check for errors, close the code file and in the terminal type “ncverilog +access+r 4bit_comparator.v”.
6. To open folder “TB” type “cd ../TB”.
7. And type “gedit tb_compare.v” to open and edit a Verilog file, add your testbench code here.
8. Close the testbench code file and type “cd ../RESULT” to open the result folder.
9. To execute the tetsbench and code type “ncverilog +access+r ../RTL/4bit_comparator.v ../TB/tb_comparator.v”.
10. This generates the vcd file in RESULT folder.
11. To check if the vcd file is present in RESULT folder enter “ls”.
12. For simulation we have used simvision.
13. To open the simulation enter simvision dump.vcd.

Code

```
// Design: 4-bit Comparator
module comparator(
   input [3:0] A, B,
   output reg greater,
   output reg equal,
   output reg less
);
always @(*) begin
```

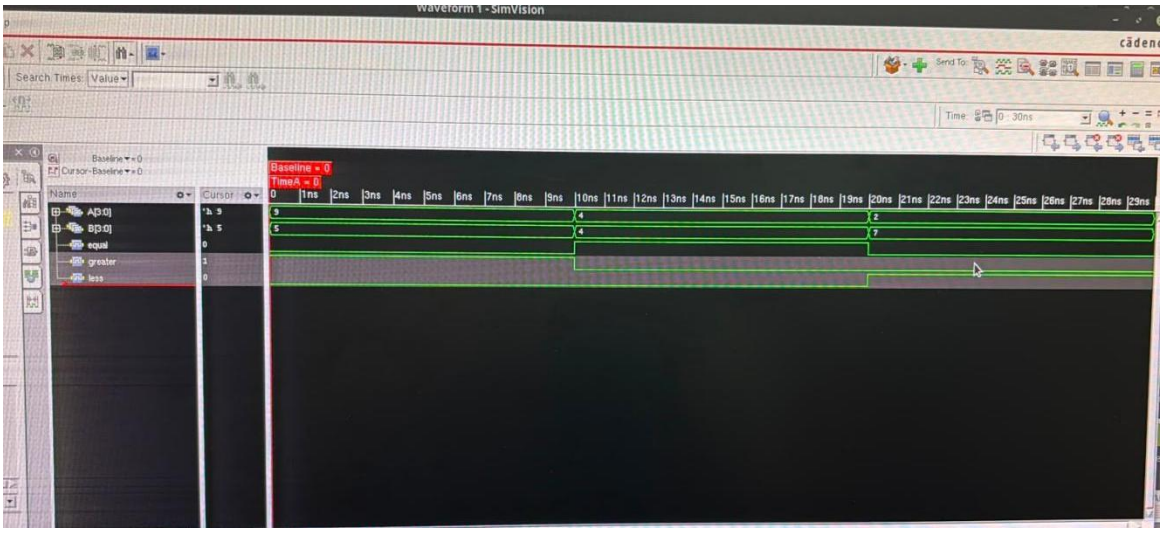
```
    greater = 0;
    equal = 0;
    less = 0;
    if (A > B)
        greater = 1;
    else if (A == B)
        equal = 1;
    else
        less = 1;
end
endmodule

// Testbench
module tb_comparator;
reg [3:0] A, B;
wire greater, equal, less;
comparator uut (

    .A(A),
    .B(B),
    .greater(greater),
    .equal(equal),
    .less(less)
);
initial begin
    $dumpfile("dump.vcd");
    $dumpvars;
    A = 4'd9; B = 4'd5; #10;
    A = 4'd4; B = 4'd4; #10;
    A = 4'd2; B = 4'd7; #10;
    $finish;
end
initial begin
    $monitor("Time=%0t A=%d B=%d Greater=%b Equal=%b Less=%b",
            $time, A, B, greater, equal, less);
```

end
endmodule

Output



SEQUENTIAL CIRCUIT

1. FOUR-BIT RING COUNTER

Theory: A 4-bit ring counter is a sequential circuit made from four flip-flops connected as a shift register, with the last output fed back to the first input. It typically circulates a single 1 through four states.

Procedure

1. Create folders with names “RTL”, “TB” and “RESULT” to store the code, testbench and simulation results respectively.
2. Open the terminal.
3. Open the folder RTL using cd command.
4. Use “gedit ring_counter.v” to open and edit a Verilog file, add your code her and save it.
5. To check for errors, close the code file and in the terminal type “ncverilog +access+r ring_counter.v”.
6. To open folder “TB” type “cd ../TB”.
7. And type “gedit tb_compare.v” to open and edit a Verilog file, add your testbench code here.
8. Close the testbench code file and type “cd ../RESULT” to open the result folder.
9. To execute the tetsbench and code type “ncverilog +access+r ../RTL/ring_counter.v ../TB/tb_ringcounter.v”.
10. This generates the vcd file in RESULT folder.
11. To check if the vcd file is present in RESULT folder enter “ls”.
12. For simulation we have used simvision.
13. To open the simulation enter simvision dump.vcd.

Code

```
// Design: 4-bit Johnson Counter
module johnson_counter(
   input clk,
   input reset,
   output reg [3:0] q
);
always @(posedge clk or posedge reset) begin
   if (reset)
```

```
        q <= 4'b0000;
    else
        q <= {q[2:0], ~q[3]};
end
endmodule

// Testbench
module tb_johnson_counter;
reg clk, reset;
wire [3:0] q;
johnson_counter uut (
    .clk(clk),
    .reset(reset),
    .q(q)
);
always #5 clk = ~clk;
initial begin
    $dumpfile("dump.vcd");
    $dumpvars;
    clk = 0;
    reset = 1;
    #10 reset = 0;
    #100;
    $finish;
end
initial begin
    $monitor("Time=%0t reset=%b q=%b", $time, reset, q);
end
endmodule
```

Output



GITHUB LINK: https://github.com/chaithrabcec25-byte/1BM25EC406_ChaithraBC